

The expected characteristic polynomial of a random lift of a
cycle:
an elementary proof, a generating function,
and a machine-checked formalization

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July 2026

Abstract

Let $\Phi_{n,r}(x)$ denote the expected characteristic polynomial of a uniformly random permutation r -lift of the cycle C_n . We give a short, self-contained derivation of the closed form $\Phi_{n,r}(x) = \chi_{C_n}(x) \cdot U_{r-1}(T_n(x/2))$, where T, U are Chebyshev polynomials and $\chi_{C_n}(x) = 2T_n(x/2) - 2$; equivalently, the d -matching polynomial of C_n (Hall–Puder–Sawin) is $U_d(T_n(x/2))$. This statement is equivalent, via the Hall–Puder–Sawin identity, to a conjecture of Hall proved combinatorially by Cochran, Groothuis, Herring, Rohatgi and Stucky (2018); our proof is independent, passing through the exponential formula and a Chebyshev generating function, and yields as byproducts the generating function $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = (1 - 2T_n(x/2)z + z^2)^{-1}$, a Fibonacci–Lucas evaluation, and a parity theorem for the quotient. The entire derivation from cycle weights to the closed form — including the exponential formula for S_r — is formalized in Lean 4 over Mathlib, with no axioms beyond Lean’s standard three. The same argument applies to every connected graph of first Betti number one, where it gives $\mu_{d,G} = V_d$ for the three-term recurrence $V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}$ in the ordinary matching polynomials of G and of the forest obtained by deleting the cycle; this answers Question 6.5 of Hall–Puder–Sawin for that family, and we show the Betti-number-one hypothesis cannot be relaxed.

1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [7], who used the $r = 2$ (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation r -lift* of a graph G replaces each vertex by r copies and each edge $\{u, v\}$ by the perfect matching $u_i \sim v_{\sigma(i)}$ of a permutation $\sigma \in S_r$ chosen independently and uniformly per edge. For the cycle C_n , gauge-fixing a spanning path reduces the lift to a single permutation $\sigma \in S_r$ on the closing edge, and the lift decomposes into disjoint cycles: each ℓ -cycle of σ contributes a cycle $C_{n\ell}$ (fixed points contribute copies of C_n). Since $\chi_{C_m}(x) =$

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$\det(xI - A_{C_m}) = 2T_m(x/2) - 2$, the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) = \mathbb{E}_{\sigma \in S_r} \prod_{\ell \in \text{cycles}(\sigma)} (2T_{n\ell}(x/2) - 2). \quad (1)$$

Theorem 1. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) = U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) = \underbrace{(2T_n(x/2) - 2)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}(T_n(x/2))}_{\Psi_r(x)}.$$

The factored form identifies the quotient Ψ_r — by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the d -matching polynomial ($d = r - 1$) — as

$$\mu_{d,C_n}(x) = U_d(T_n(x/2)),$$

which is equivalent (through $\mathcal{P}_d(2t) = U_d(t)$, $\mathcal{C}_n(2t) = 2T_n(t)$ and the composition identity $U_{dn+n-1} = U_d(T_n)U_{n-1}$) to *Hall's conjecture* $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$, proved in [3]. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

Corollary 2. $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = \frac{1}{1 - 2T_n(x/2)z + z^2}, \quad \text{and} \quad \sum_{r \geq 0} \Phi_{n,r}(x) z^r = \frac{(1 - z)^2}{1 - 2T_n(x/2)z + z^2}.$

Corollary 3 (real-rootedness). *All roots of $\Phi_{n,r}$ lie in $[-2, 2]$: with $x = 2 \cos \theta$, χ_{C_n} vanishes at $\theta = 2\pi m/n$ and $\Psi_r = \sin(rn\theta)/\sin(n\theta)$ vanishes at $\theta = m\pi/(rn)$ with $r \nmid m$ (when $r \mid m$ the denominator vanishes too and $\Psi_r = \pm r \neq 0$); this leaves exactly $n(r - 1)$ roots, matching $\deg \Psi_r$.*

Corollary 4 (Fibonacci–Lucas evaluation). $\Phi_{n,r}(3) = (L_{2n} - 2) F_{2nr}/F_{2n}$, since $T_n(3/2) = L_{2n}/2$ and $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$.

Corollary 5 (parity). $\Psi_r(-x) = (-1)^{n(r-1)}\Psi_r(x)$: the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite C_n it is an even polynomial for every r ; at $r = 2$ it is the classical matching polynomial $2T_n(x/2)$ of C_n , consistent with *Godsil–Gutman* [1].)

2 Proof

Write $f(\ell) = 2T_\ell(Y) - 2$ with $Y = T_n(x/2)$; by the composition identity $T_{n\ell} = T_\ell \circ T_n$ this is the per-cycle factor in (1).

Step 1 (exponential formula). For any commutative ring R and $f : \mathbb{N} \rightarrow R$, the S_r -average $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \text{cycles}(\sigma)} f(\ell)$ satisfies

$$r \Phi_r = \sum_{k=1}^r f(k) \Phi_{r-k}, \quad \Phi_0 = 1, \quad (2)$$

the coefficient form of $\sum_r \Phi_r z^r = \exp(\sum_\ell f(\ell) z^\ell / \ell)$. (Proof: group σ by cycle type λ with Cauchy's count $r!/z_\lambda$, mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

Step 2 (Chebyshev generating function). With $f(\ell) = 2T_\ell(Y) - 2$,

$$\sum_{\ell \geq 1} f(\ell) \frac{z^\ell}{\ell} = -\log(1 - 2Yz + z^2) + 2\log(1 - z) \implies \sum_{r \geq 0} \Phi_r z^r = \frac{(1 - z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against $\sum U_r(Y) z^r = (1 - 2Yz + z^2)^{-1}$ gives $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$, and the three-term recurrence $U_r + U_{r-2} = 2YU_{r-1}$ collapses this to $(2Y - 2)U_{r-1}(Y)$. $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with $\Phi_0 = 1$ determines Φ over a characteristic-zero domain.

3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because C_n has $|E| - |V| + 1 = 1$. The argument therefore applies verbatim to any connected G with $b_1(G) = |E| - |V| + 1 = 1$, a cycle with trees attached, and we record what it gives.

By [2] the expected characteristic polynomial of a random r -lift of *any* graph factors as $\chi_G \cdot \mu_{r-1,G}$; the content below is therefore not that factorization but a closed form for the factor $\mu_{d,G}$ itself. Question 6.5 of [2] asks which parts of the theory of the matching polynomial $\mu_{1,G}$ generalize to $\mu_{d,G}$; the following answers it for $b_1(G) = 1$.

Let C be the unique cycle of G and let $G - V(C)$ be the forest obtained by deleting its vertices. Write μ_H for the ordinary matching polynomial of a graph H , with $\mu_\emptyset = 1$. Define

$$V_0 = 1, \quad V_1 = \mu_G, \quad V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}. \quad (3)$$

Theorem 6. *For every connected G with $b_1(G) = 1$ and every $d \geq 0$, $\mu_{d,G} = V_d$.*

For $G = C_n$ the deleted forest is empty, so $\mu_{G-V(C)} = 1$ and (3) is the recurrence for $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$: Theorem 1 is the case $\mu_{G-V(C)} = 1$. In general the attached trees enter only through $\mu_{G-V(C)}$; for the triangle with one pendant edge, $\mu_G = x^4 - 4x^2 + 1$ and $\mu_{G-V(C)} = x$.

The proof goes through the twisted characteristic polynomial. Let $e = uv$ be the unique non-tree edge, let $A_G(z)$ be the adjacency matrix of G with the (u, v) entry replaced by z and the (v, u) entry by z^{-1} , and set $F(x, z) = \det(xI - A_G(z))$.

Lemma 7. $F(x, z) = \mu_G(x) - \mu_{G-V(C)}(x)(z + z^{-1})$.

Proof. By the Sachs coefficient theorem $\det(xI - A)$ expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and $A_G(z)^\top = A_G(z^{-1})$, the determinant is of the form $P(x) + Q(x)(z + z^{-1})$. Replacing z by ± 1 and averaging kills exactly the elementary subgraphs whose cycle traverses e ; as G has the single cycle C , those are the subgraphs containing C , so $P = \mu_G$, and the surviving terms contribute $Q = -\mu_{G-V(C)}$, a cycle component carrying the Sachs weight -2 . $\square$

Proof of Theorem 6. Gauge-fix a spanning tree: the r -lift is determined by one permutation $\sigma \in S_r$ on e , and its components are indexed by the cycles of σ , the component of an ℓ -cycle being the ℓ -fold cyclic cover G_ℓ . Decomposing the $\mathbb{Z}/\ell$ action into characters gives the classical factorization $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$ with $\omega = e^{2\pi i/\ell}$ [5, 6]. With P, Q as in Lemma 7 and $y = -P/(2Q)$,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1} \left(P + 2Q \cos \frac{2\pi j}{\ell} \right) = (-Q)^\ell (2T_\ell(y) - 2),$$

so by the exponential formula (2), with $w = -Qz$,

$$\sum_{r \geq 0} \Phi_{G,r} z^r = \exp \left(\sum_{\ell \geq 1} (2T_\ell(y) - 2) \frac{w^\ell}{\ell} \right) = \frac{(1-w)^2}{1-2yw+w^2},$$

and extracting coefficients as in Section 2 gives $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1} U_{r-1}(y)$. Comparing with $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ [2] yields $\mu_{d,G} = (-Q)^d U_d(y)$, which satisfies (3). $\square$

Since $U_d(y) = 2^d \prod_{k=1}^d (y - \cos \frac{k\pi}{d+1})$, the theorem can be restated as a product, $\mu_{d,G}(x) = \prod_{k=1}^d F(x, e^{ik\pi/(d+1)})$, whence:

Corollary 8. *For $b_1(G) = 1$, $\mu_{d,G}$ is a product of d real-rooted polynomials, and its roots are exactly the eigenvalues of the Hermitian matrices $A_G(e^{ik\pi/(d+1)})$, $1 \leq k \leq d$.*

This is the real-rootedness theorem of [2] for this family, with the roots identified.

The roots can be located much more precisely than that. Since $b_1(G) = 1$ the group $\pi_1(G)$ is $\mathbb{Z}$, so the universal cover T of G is the $\mathbb{Z}$ -cover, and T is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres $A_G(e^{i\theta})$. Hence

$$\text{spec}(T) = \bigcup_{\theta} \text{spec} A_G(e^{i\theta}) = \left\{ x : \exists s \in [-1, 1], \mu_G(x) = 2s \mu_{G-V(C)}(x) \right\},$$

a finite union of bands, together with a flat band at each common zero of μ_G and $\mu_{G-V(C)}$, where $F(x, \cdot)$ vanishes identically.

Corollary 9. *For $b_1(G) = 1$ the roots of $\mu_{d,G}$ are exactly the solutions of $\mu_G(x) = 2 \cos(\frac{k\pi}{d+1}) \mu_{G-V(C)}(x)$ for $1 \leq k \leq d$. In particular every root of every $\mu_{d,G}$ lies in $\text{spec}(T)$, and as $d \rightarrow \infty$ the roots equidistribute on $\text{spec}(T)$ with respect to its density of states.*

Proof. Immediate from Theorem 6 in the form $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$ and $F(x, e^{i\theta}) = \mu_G - 2 \cos \theta \mu_{G-V(C)}$. The equidistribution is the arcsine law for $\cos(k\pi/(d+1))$ transported through the fibre map. $\square$

Hall–Puder–Sawin place the roots of $\mu_{d,G}$ in the Ramanujan *interval* $[-\rho, \rho]$, ρ being the spectral radius of T . Corollary 9 places them in $\text{spec}(T)$, which is a union of bands and is a proper subset of that interval whenever T has a spectral gap: no root of any $\mu_{d,G}$ can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where $\mu_{G-V(C)} = 1$ and $\text{spec}(T) = [-2, 2]$ is the whole

interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of $[-\rho, \rho]$ that is a gap):

G	ρ	gap proportion
C_n	2	0%
triangle + pendant	2.170	36.5%
C_4 + pendant	2.136	45.9%
triangle + P_2	2.214	51.3%
C_4 + P_3	2.189	54.1%
triangle + $K_{1,3}$	2.514	65.9%

So the cycle is not merely the easiest case of Theorem 6: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

A conjecture for all graphs. Corollary 9 is proved by Floquet theory, which is available only because $\pi_1(G) = \mathbb{Z}$. Its *statement*, however, makes sense for every graph, and we know two cases of it. At $d = 1$ it is classical: $\mu_{1,G}$ is the ordinary matching polynomial, whose roots are eigenvalues of the path tree of G and hence lie in $\text{spec}(T)$ (Godsil). At $b_1(G) = 1$ it is Corollary 9, for every d . We therefore propose:

Conjecture 10. *For every finite graph G and every $d \geq 1$, all roots of $\mu_{d,G}$ lie in $\text{spec}(T)$, the spectrum of the universal cover of G .*

This strengthens the root-localization theorem of [2], which places the roots in the interval $[-\rho, \rho]$ with ρ the spectral radius of T ; the conjecture places them in $\text{spec}(T)$ itself. The two agree whenever T has no spectral gap — in particular for regular G , where T is a regular tree and $\text{spec}(T) = [-2\sqrt{q}, 2\sqrt{q}]$ is an interval. They differ exactly when T has a gap, and the smallest examples are biregular: for $K_{2,3}$ the universal cover is the $(3, 2)$ -biregular tree, with spectrum $\{0\} \cup \pm[\sqrt{2} - 1, \sqrt{2} + 1]$ and hence a gap $(0, \sqrt{2} - 1)$ inside the Ramanujan interval. We have computed $\mu_{d,K_{2,3}}$ for $d \leq 4$, and $\mu_{d,K_{2,4}}$ for $d \leq 3$ (universal cover the $(4, 2)$ -biregular tree, gap $(0, \sqrt{3} - 1)$), by averaging matching polynomials over all d -covers: in every case no root lies in the gap. Note that $b_1(K_{2,3}) = 2$ and $b_1(K_{2,4}) = 3$, so these are outside the reach of Corollary 9.

A proof of Conjecture 10 cannot go through Floquet theory, since for $b_1 \geq 2$ the relevant cover is a tree with free deck group rather than a periodic chain. What the $b_1 = 1$ case does show is that the interval in [2] is not the natural receptacle for the roots, and that the correct one is spectral rather than metric.

Related work. Two ingredients are classical and are used, not claimed. The factorization of the characteristic polynomial of a cyclic cover into twisted determinants at roots of unity is the block-circulant diagonalization [5, 6]; in the infinite-cover limit it is the Floquet–Bloch decomposition of a periodic discrete operator, and $y = -P/(2Q)$ is the corresponding Floquet discriminant. The identity $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ is [2]. Closest in form is McSorley and Feinsilver [4]. For a path whose edges are cyclically labelled with t labels they establish a (τ, Δ) -recurrence $\Phi_N = \tau\Phi_{N-1} - \Delta\Phi_{N-2}$, which is the algebraic skeleton of (3), and their Theorem 4.4 solves it as $\Delta^{N/2}U_N(\tau/2\sqrt{\Delta})$, exactly the form used here. In the univariate

specialization their τ is the matching polynomial of the cycle C_t and $\Delta = (-1)^t x_1 \cdots x_t$ is a monomial; passing between their counting normalization and the characteristic one used here is the homogenization $V_d \mapsto c^d V_d$, under which a monomial Δ becomes $\mu_{G-V(C)} = 1$. Their Theorems 4.4 and 4.5 together give $U_{dn+n-1} = U_d(T_n)U_{n-1}$, the identity invoked in Section 1 to pass between Theorem 1 and Hall’s conjecture, so that bridge was already available in 2009. Their parameter is the number of label periods along a path and their graphs are paths, cycles and rooted trees; there is no covering and no cycle with trees attached. What is added here is the pair $(\mu_G, \mu_{G-V(C)})$, whose second entry is a genuine polynomial recording the attached trees rather than a monomial, together with the covering interpretation that makes the recurrence a statement about $\mu_{d,G}$.

The boundary is sharp. For $b_1(G) = b$ the lift is determined by b permutations, that is by an action of the free group F_b on $[r]$, whose orbits are the components of the lift. The exponential formula still applies, in the form $N_r = \sum_{\ell} \binom{r-1}{\ell-1} C_{\ell} N_{r-\ell}$ with $N_r = (r!)^b \Phi_{G,r}$ and C_{ℓ} the sum of χ over the transitive b -tuples on $[\ell]$; we have verified this for $b \leq 3$. What fails is the atom. For $b = 1$ every transitive tuple gives the *same* cover G_{ℓ} , so $C_{\ell} = (\ell - 1)! \chi_{G_{\ell}}$: one graph per ℓ , with $\chi_{G_{\ell}}$ Chebyshev in ℓ . For $b \geq 2$ the transitive tuples number $(\ell - 1)!$ times the count of index- ℓ subgroups of F_b and spread over many non-isomorphic covers with distinct spectra, so C_{ℓ} carries no such structure. Accordingly Theorem 6 is not merely unproved for $b \geq 2$: it is false there already at $d = 1$, as we checked for the theta graph, the bowtie, K_4 , $K_{2,3}$ and two triangles joined by an edge.

Theorem 6 was verified by exact integer computation against the definition of $\mu_{d,G}$ as an average of matching polynomials over all d -covers, and against brute-forced lift averages, on thirteen named unicyclic graphs for $r \leq 6$ and on a random sample of thirty-four for $r \leq 5$; Lemma 7 was checked on eleven.

4 The formalization

The core algebraic derivation — the exponential formula, the closed form of Theorem 1 for all n, r (equivalently, everything from the cycle weights (1) to the closed form), its factorization, and the parity Corollary 5 — is machine-checked in Lean 4 over Mathlib (v4.30). The main entry points:

- `Paper2.expFormula` — the exponential formula (2) in integer form $r A_r = \sum_k r^{(k)} f(k) A_{r-k}$ over any commutative ring, proved via Mathlib’s Cauchy cycle-type count (`Equiv.Perm.card_of_cycleType`);
- `Paper2.thm1_Sr` — Theorem 1 in the form: the expected cycle weight (1) equals the closed form, for all n, r , over any $\mathbb{Q}$ -algebra domain;
- `Paper2.cG_factored`, `Paper2.quotient_parity` — the factorization and Corollary 5.

The three remaining corollaries are formalized only in part: Corollary 2’s generating function appears as the finite recurrence (2) rather than a literal power-series identity; of Corollary 3 only the underlying trigonometric factorization (`cor1_factored`) is checked, not the “all roots in $[-2, 2]$ ” statement end-to-end; and Corollary 4 (`cor2_phi3`) is a bounded `native_decide` over $n, r \leq 6$.

Trust base for the general theorems: `propext`, `Classical.choice`, `Quot.sound` only (checked with `#print axioms`); the bounded `native_decide` checks additionally invoke `Lean.ofReduceBool`. No `sorry`. The single remaining unformalized ingredient of the *graph-theoretic* statement is the sentence before (1): a lift of a cycle is a disjoint union of cycles — elementary covering-space combinatorics for which Mathlib currently lacks the vocabulary.

Remark 11. Everything here is stated for the uniform permutation model; by [2] the quotient coincides with the $(r - 1)$ -matching polynomial, so Theorem 1 can alternatively be deduced from [3]. We believe the generating function of Corollary 2, the evaluation of Corollary 4, and the formalization are new.

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